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References

COMPARISON OF THE EFFICACY OF NARATRIPTAN AND SUMATRIPTAN FOR ACUTE MIGRAINE TREATMENT: A COMPARATIVE STUDY

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Background and Aims

Migraine is a prevalent and disabling neurological disorder that negatively affects quality of life, productivity, and healthcare costs. Diagnosis follows the ICHD-3 criteria, and management includes acute and preventive treatments. Triptans are the drugs of choice for the acute treatment of migraine attacks due to their selective action on the trigeminovascular system. Among the available triptans, sumatriptan (SUM) is the most widely used worldwide, whereas naratriptan (NAR), with its longer half-life and slower onset of action, represents an alternative for specific patient profiles. This study compares the efficacy of SUM and NAR in the acute treatment of migraine.

Objective

To compare the efficacy of sumatriptan and naratriptan in the acute treatment of migraine based on medical record data.

Methods

This prospective study included seven patients diagnosed with migraine who were recruited after signing an informed consent form and followed up, on average, every two months, for a minimum of six months. Patients using SUM or NAR as definitive acute rescue medication were included. The variables analyzed included sex, age, medication used, and time to reduction in migraine attacks. Patient characteristics were described using percentage distributions and frequency distributions.

Quantitative variables were summarized using minimum, maximum, mean, median, standard deviation, and 95% confidence intervals. The normality of continuous variables was assessed using the Shapiro–Wilk test when applicable ($n \geq 3$). Response time was compared between groups using the Mann–Whitney test because the SUM group did not follow a normal distribution and the NAR group had an insufficient sample size. Analyses were performed using Epi Info version 7.2.7, with a significance level of 5% and Bonferroni correction for multiple comparisons ($\alpha = 0.025$).

Results

Seven patients were analyzed, all female, with a mean age of 39.1 ± 12.0 years. The SUM group (n = 5) had a mean age of 39.0 ± 14.7 years and a mean response time of 2.4 ± 2.2 months (median 1.0; minimum 1; maximum 6; 95% CI: −0.32–5.12). The NAR group (n = 2) had a mean age of 39.5 ± 0.7 years and a mean response time of 1.0 ± 0.0 months (median 1.0; minimum 1; maximum 1; 95% CI not calculable due to zero standard deviation). Both triptans achieved resolution of migraine attacks in 100% of cases in which they were maintained as definitive medication. Notably, one patient initially received NAR without a satisfactory response and subsequently achieved attack control after switching to SUM, highlighting the individual variability in response to triptans. The Shapiro–Wilk test indicated a non-normal distribution in the SUM group ($W = 0.754$; $p = 0.033$) and was not applicable to the NAR group (n = 2), supporting the use of the Mann–Whitney test for between-group comparisons. The comparison of response times did not reach statistical significance after Bonferroni correction ($U = 7.0$; $p = 0.469$; corrected $\alpha = 0.025$).

Conclusions

Both triptans demonstrated efficacy in controlling migraine attacks in the cases analyzed, with resolution achieved in 100% of cases in which they were maintained as definitive medication. NAR showed a shorter response time (1.0 ± 0.0 months) than SUM (2.4 ± 2.2 months); however, this difference was not statistically significant after Bonferroni correction ($p = 0.469$; corrected $\alpha = 0.025$). The small sample size was the main limitation of the analysis. The case of treatment failure with NAR followed by a response to SUM illustrates that a lack of response to one triptan does not predict treatment failure with other triptans, a finding consistent with the literature. The exclusively female sample is consistent with the higher prevalence of migraine among women. The choice between triptans should consider their individual pharmacological properties and each patient's clinical response.